Rishi Rajesh Verma

rishiv989 | rishiverma2004 | rishiv989@gmail.com | +91 9648624154

Summary

Aspiring AI/ML Engineer and Data Science professional with knowledge of Python, machine learning models, data visualization, and statistical analysis, looking to leverage analytical and problem-solving skills to create scalable AI solutions.

Skills

Languages	Python, SQL
Libraries/Frameworks	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, FastAPI.
ML/AI	Machine Learning, Deep Learning, NLP, RAG (Retrieval-Augmented Generation), Semantic Search, Explainable AI (SHAP, LIME), Computer Vision (YOLOv8, MediaPipe), ASR (Whisper)
DBMS & Storage	MySQL, PostgreSQL, Vector Databases (FAISS, Pinecone), Amazon S3
Tools	Git, GitHub, VS Code, Jupyter Notebook, Google Colab, MySQL Workbench, MS Excel, Ollama, Docker

Work Experience

Thynktech India - Associate Software Engineer Sep 2025 - present

- Worked on developing and refining machine learning models for ThynkLearn, an AI-powered EdTech platform with a personalized learning focus.
- Using a RAG architecture, a chatbot and automated test generator were created, utilizing AWS S3 for effective content storage and retrieval and PostgreSQL for session management.
- Semantic search and OpenAI GPT models were used to process textbook data and produce pertinent, syllabus-based assessments.
- Developed a real-time proctoring system that uses MediaPipe Face Mesh to track eye movement, YOLOv8 to detect unwanted items, and DeepFace to verify identity.
- Fuzzy matching algorithms and speech recognition were combined to assess pupils' reading fluency.
- In order to comprehend and clarify image-based homework issues, I worked using vision-language models (Ollama).

Technohacks Pvt Ltd - Data Analyst Intern Dec 2024 - Jan 2025

- Analysing large datasets to derive actionable insights for clients.
- Visualizing data insights using Python libraries like Matplotlib and Seaborn.
- Created Dashboard Using PowerBI.

Projects

RAG-Based AI Teaching Assistant Link

- Developed a fully localized Retrieval-Augmented Generation (RAG) based AI Teaching Assistant for a video-based course, enabling students to interactively query and learn from course content.
- Built an end-to-end data pipeline that processes video lectures by extracting audio, generating transcriptions using Faster-Whisper, and converting them into vector embeddings for semantic search.
- Implemented a FastAPI-based backend with endpoints for question answering, quiz generation, and dynamic video ingestion, leveraging cosine similarity to retrieve relevant content chunks.
- Integrated local LLMs via Ollama (LLaMA 3.2 and BGE-M3) for context-aware response generation and embedding creation, eliminating dependency on external APIs.
- Added multi-language support (English, Hindi, Marathi), intent-based responses, and features like timestamped video navigation, making the system an efficient, scalable, and interactive learning assistant.

Explainable AI For Aircraft Failure Diagnosis And Decision Support System Link

- Developed an Explainable AI (XAI)-driven predictive maintenance system for aircraft engine health monitoring using the NASA CMAPSS dataset.
- Built end-to-end machine learning pipelines to predict Remaining Useful Life (RUL) and classify engine health states (Normal, Warning, Critical) using advanced models such as XGBoost and AutoGluon.
- Integrated SHAP and LIME to provide both global and local interpretability, enabling transparent analysis of feature importance and individual predictions.
- Designed an interactive Streamlit dashboard to visualize predictions, model insights, and explanation outputs, along with a Three.js-based 3D digital twin interface that maps real-time engine health indicators and anomalies onto a rotating jet engine model.
- The system delivers accurate, interpretable, and visually intuitive insights to support reliable maintenance decision-making.

Education

2022 - present	B.E. in Artificial Intelligence and Data Science - **SPPU**	(GPA: 8.0/10.0)
2022	Class 12th - **MSBSHSE**	(64 %)
202	Class 10th - **MSBSHSE**	(78 %)

Publications & Certifications

- Prof. Snehal Javheri and Mr. Rishi Verma (2025). "Paper: Enhancing Aircraft Safety through Explainable AI-Based Failure Detection Systems" Link
- Google AI Workshop (Google, 2025)
- Data Engineering(Geeks For Geeks, 2025)